

Доказательство Леммы Ньютона-Лейб-

$$\int_a^b f(x) dx = F(b) - F(a)$$

$$(T_1) \quad] f(x) \in C^0[a, b]]$$

$$] F(x) : F'(x) = f(x), x \in [a, b]$$

$$\Rightarrow \int_a^b f'(x) dx = F(b) - F(a) = F(x) \Big|_a^b$$

$$\Rightarrow] f(x) \in C^0[a, b] \Rightarrow] F(x) = \int_a^x f(t) dt + C + C, \text{ где } F(a) = \int_a^a f(t) dt + C = 0 + C \Rightarrow C = F(a)$$

$$F(x) = \int_a^x f(t) dt + F(a),$$

$$\Rightarrow F(b) = \int_a^b f(t) dt + F(a)$$

$$\Rightarrow \int_a^b f(t) dt = F(b) - F(a)$$

$$(T_2) \quad] f \in R[a, b], F(x) : F'(x) = f(x) \sim \text{непр.}$$

$$\Rightarrow \int_a^b f dx = F(b) - F(a)$$

$$\Delta \quad \textcircled{2} \quad F(b) - F(a)$$

$$\textcircled{2} \quad \tau: x_i = a + \frac{b-a}{n} i, \quad i = 0, n$$

$] f - \text{непр. } f \text{ на } [a, b]$

$$\begin{aligned} F(b) - F(a) &= F(x_n) - F(x_0) = [F(x_n) - F(x_{n-1})] \\ &+ [F(x_{n-1}) - F(x_{n-2})] + [F(x_{n-2}) - F(x_{n-3})] + \dots \\ &+ [F(x_1) - F(x_0)] \stackrel{\text{Трап. н.} = F'(\xi_i)}{=} \sum_{i=1}^n f(\xi_i) \Delta x_i \end{aligned}$$

$$\forall (\tau^n, \xi^n), f \in R[a, b], \text{ при } \lambda(\tau^n) \rightarrow 0 \Rightarrow \lim_{\lambda(\tau) \rightarrow 0} S_\tau(f, \xi) = \int_a^b f(x) dx,$$

то с гр. с.

$$F(b) - F(a) = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x_i = \int_a^b f(x) dx$$

$\textcircled{I}$ Формула интегрирования по частям.

$$] u(x), v(x) \in D[a, b]$$

$$u'(x), v'(x) \in R[a, b] \quad | \quad \Rightarrow$$

$$\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$$

$$\Delta \quad uv', \quad v u' \in R[a, b]$$

$$\textcircled{2} \quad (u \cdot v)' = u' \cdot v + v' \cdot u \in R[a, b]$$

$$\Rightarrow \int_a^b u' v dx + \int_a^b v' u dx = \int_a^b (u \cdot v)' dx = (u \cdot v) \Big|_a^b$$

$$\Rightarrow \int_a^b u dv = - \int_a^b v du + (u \cdot v) \Big|_a^b$$

$\textcircled{I}$ Формула замены переменных.

$$] f \in C^0[a, b], x = \varphi(t): [\alpha, \beta] \rightarrow [a, b]$$

$$\varphi(t) \in D[\alpha, \beta], \varphi'(t) \in R[\alpha, \beta]$$

$$\Rightarrow \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x) dx = \int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) dt.$$

$$\Delta \underbrace{f(\varphi(t))}_{\in C^0[\alpha, \beta]} \varphi'(t) \in R[\alpha, \beta]. \quad \text{T.k. } F(x) - \text{непрерывно } f(x)$$

$$\Rightarrow F(\varphi(t)) - \text{непрерывно, где } f(\varphi(t)) \varphi'(t) \in R[\alpha, \beta]$$

$$\left\{ F'_z(\varphi(t)) = f(\varphi(t)) \cdot \varphi'(t) \right\} \xrightarrow{\text{ФН-Анз}} \frac{\varphi(\beta) - \varphi(\alpha)}{\beta - \alpha}$$

$$\Rightarrow \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x) dx = F(\varphi(t)) \Big|_{\alpha}^{\beta} = F(\varphi(\beta)) - F(\varphi(\alpha)) =$$

$$= \int_{\varphi(\alpha)}^{\varphi(\beta)} f(x) dx = \int_{\alpha}^{\beta} f(x) dx \quad \Delta$$

$$\text{Typ } \int_0^a \sqrt{a^2 - x^2} dx = \begin{cases} x = a \sin t \\ dx = a \cos t dt \\ x=0 \Rightarrow t=0 \\ x=a \Rightarrow t=\frac{\pi}{2} \end{cases} \left\{ \begin{aligned} &= a^2 \int_0^{\pi/2} \cos^2 t dt = \frac{a^2}{2} \int_0^{\pi/2} (1 + \cos 2t) dt = \\ &= \frac{a^2}{2} \left(\frac{\pi}{2} + \frac{1}{2} \sin 2t \Big|_0^{\pi/2} \right) = \frac{a^2}{2} \left(\frac{\pi}{2} + 0 \right) = \frac{\pi a^2}{4} \end{aligned} \right.$$

$$= \frac{a^2}{2} \left(\frac{\pi}{2} + \frac{1}{2} \sin 2t \Big|_0^{\pi/2} \right) = \frac{a^2}{2} \left(\frac{\pi}{2} + 0 \right) = \frac{\pi a^2}{4}$$

$$\square \quad \exists f \in R[0, a], \quad f - \text{четная} \Rightarrow$$

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

$$\Delta \quad \exists f(x) \in R[0, a]. \quad \text{T.k. } f(-x) = f(x) \Rightarrow f(x) \in [-a, a]$$

$$\Rightarrow \int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx \quad (\equiv)$$

$$\Rightarrow \int_{-a}^0 f(x) dx = \begin{cases} x = -t \\ dx = -dt \\ x = -a \Rightarrow t = a \\ x = 0 \Rightarrow t = 0 \end{cases} \left\{ \begin{aligned} &= - \int_a^0 f(-t) dt = \\ &= \int_0^a f(t) dt = \int_0^a f(x) dx; \end{aligned} \right.$$

$$= \int_0^a f(t) dt = \int_0^a f(x) dx;$$

$$(\equiv) 2 \int_0^a f(x) dx \quad \Delta$$

$$\textcircled{T} \quad] f \in R[-a, a], \quad f - \text{ker } T, \Rightarrow$$

$$\int_{-a}^a f(x) dx = 0$$

$$\Delta \quad T.k. \quad f(-x) = -f(x) \quad \forall x \in \mathbb{C}$$

$$f(-x) = -f(x), \quad x \in [-a, a] \Rightarrow f(x) \in R[-a, a]$$

$$\Rightarrow \int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx =$$

$$= \left\{ \begin{array}{l} x = -t \end{array} \right\} = \int_0^a f(-x) dx + \int_0^a f(x) dx =$$

$$= \int_0^a [f(-x) + f(x)] dx = \int_0^a (-f(x) + f(x)) dx =$$

$$= 0 \quad \Delta$$

$$\text{Auf} \quad \int_{-\pi/2}^{\pi/2} \frac{\sin^5 x \cdot \cos^8 x}{1+x^2} dx = 0$$

$$\textcircled{T} \quad] f \in R[0, T], \quad T\text{-periodisch: } f(x+T) = f(x)$$

$$\Rightarrow \int_a^{a+T} f(x) dx = \int_0^T f(x) dx, \quad \forall a \in \mathbb{R}$$

$$\Delta \quad \int_a^{a+T} f(x) dx = \int_a^0 f(x) dx + \int_0^T f(x) dx + \int_T^{a+T} f(x) dx =$$

$$= \left\{ \begin{array}{l} x = t + T \end{array} \right\} = \int_a^0 f(x) dx + \int_0^T f(x) dx +$$

$$+ \int_0^a f(t+T) dt = \int_0^a f(t) dt = \int_0^a f(x) dx \quad \Delta$$

$$\text{Auf} \quad \int_{-\pi/2}^{5\pi/2} \sin^5 x \cdot \cos^8 x dx = \int_{-\pi/2}^{\pi/2} + \int_{\pi/2}^{5\pi/2} = \int_0^{2\pi} = \int_{-\pi}^{\pi} = 0$$

Геометрические приложения

опр] $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$

Величина $|x| = \sqrt{\sum_{i=1}^n x_i^2}$ наз. расстоянием от: Т. x до начала к-т.

опр. Отображение $V: \mathbb{R}^n \rightarrow \mathbb{R}^n$ наз. изометризмом, если $|x - y| = |V(x) - V(y)|$

опр Функция меры

$$S: \mathcal{U} \rightarrow \mathbb{R}$$

заданная на сем-ве квоцируемых н/м-ф, наз. мерой, если:

① $S(A) \geq 0, A \in \mathcal{U}$

② если $A, B \in \mathcal{U}, A \cap B = \emptyset$, то $A \cup B \in \mathcal{U}$
и $S(A \cup B) = S(A) + S(B)$

③ $S_{\text{нр-ка}} = a \cdot b$

④ если $A \in \mathcal{U}, V$ -гв., то $V(A) \in \mathcal{U}$ и
 $S(V(A)) = S(A)$.